Md. Akram Khan

Dhaka, Bangladesh

(+880) 1612304791 | thankyouakramkhan@gmail.com | md-akram-khan.github.io | Md-Akram-Khan | md-akram-khan-534363291

Education

Department of Computer Science and Engineering, University of Dhaka — *Dhaka, Bangladesh*

Bachelor of Science (B.Sc.), CGPA: x.xx/4.00 — *2023 - Present*

Chittagong College — *Chittagong, Bangladesh*

Higher Secondary Certificate, GPA: 5.00/5.00 — *2022*

Chattogram Collegiate School — *Chittagong, Bangladesh*

Secondary School Certificate, GPA: 5.00/5.00 — *2020*

Experience

University of Dhaka — *Dhaka, Bangladesh*

Research Collaborator, Department of Computer Science and Engineering — *April 2026 - Present*

Research under the supervision of **Dr. Md. Mosaddek Khan**, Associate Professor, Department of Computer Science and Engineering, University of Dhaka.

Udvash Academic & Admission Care — *Dhaka, Bangladesh*

Math & ICT Content Developer — *2023 - Present*

Creating educational content and simplifying complex mathematical and ICT concepts for academic learners.

- Developed **100+ practice problems** with detailed solution guides for mathematics and ICT.
- Designed visual aids and interactive content that improved student comprehension.
- Reviewed and updated existing study materials, collaborating with educators to improve content quality.

Projects

Electricity Billing System (Github Link) — *Java, Swing, PostgreSQL, Supabase, Python*

Object Oriented Programming Lab Project — *2025*

A Java Swing desktop app for an electricity provider: admins manage customers, meters and monthly bills; customers view and pay them. Implements the BERC slab tariff with lifeline rate, demand charge, 5% VAT and late fees; bills generated from a meter photo with a checker that flags suspicious bills; online payment through SSLCommerz (bKash, Nagad, Rocket and cards).

Cognify (Github Link) | (Live Demo) — *FastAPI, React 19, MongoDB, Tailwind CSS, Groq, Gemini*

AI-Powered Learning Platform · Team Project

A full-stack study platform with an AI tutor (Groq as the primary model, Gemini as the fallback), practice exams generated on any topic in English, Bengali, Hindi, Spanish and German, and summaries and quizzes extracted from uploaded PDFs and slide decks. Includes an offline demo mode that runs without API keys or a database.

StockHub (Github Link) | (Live Demo) — *FastAPI, React, Docker, Nginx, Tailwind CSS*

Warehouse and Storage Management · Team Project

A warehouse management system with separate workspaces for customers, employees and admins, covering storage applications, inventory tracking, branch capacity and audit logs. Uses role-based access control and optimistic locking for concurrent inventory updates; containerised with Docker and served through Nginx.

DrugScript (Github Link) *Flutter, Dart, Firebase*

Cross-Platform Healthcare App · Team Project

A healthcare app bringing prescriptions, medicine and ambulances together: digital prescriptions shared and scanned as QR codes, medicine search and delivery flows for patients, shops and delivery staff, an ambulance service hub for patients and drivers, and medication reminders.

Chattogram Emergency Routing (Github Link) *JavaScript, HTML, CSS, Google Maps, Chart.js*

AI Routing for Ambulances · Personal Project *2026*

A dispatch and routing simulator for ambulances and rescue teams across Chattogram, modelling monsoon flooding, port traffic, landslide risk and night-time safety over 45 metropolitan nodes and 60+ road links. Edge costs change with road type, traffic, flood depth and time of day. A* with a Haversine heuristic is benchmarked against BFS, DFS, UCS, IDS and Greedy search.

CatVision (Github Link) | (Live Demo) *Python, NumPy, Flask, React, Vite*

Machine Learning from First Principles · Personal Project *2026*

A cat vs non-cat image classifier. The model is logistic regression implemented from scratch in NumPy with no ML framework, served by a Flask API on Render, with a React + Vite frontend deployed to GitHub Pages.

Skills

Programming Languages	Python, C++, Java, JavaScript, TypeScript, Dart
Web Development	React, Vite, Tailwind CSS, HTML & CSS, FastAPI, Flask
AI & Data	NumPy, Logistic Regression, Graph Search (A*, UCS, BFS), Groq & Gemini APIs
Mobile	Flutter, Firebase
Database	PostgreSQL, Supabase, MongoDB
Tools & Deployment	Git & GitHub, Docker, Nginx, Vercel, Render, GitHub Pages

Problem Solving

Solved **840+ problems** across online judges, with C++ practice notebooks on Number Theory, Data Structures & Algorithms, Graph Theory and the CSES Problem Set.

Codeforces	Thank_you · 575 problems solved
Virtual Judge	Thank_youu · 175 problems solved
LeetCode	Thank_you · 67 problems solved
GeeksforGeeks	thankyou05 · 24 problems solved

Achievement

December, 2025	**NahamCon CTF 2025**, Ranked **74th of 859 teams** (international) as team Restless_goblins, scoring 1,326 points and solving 8 of 25 challenges among 1,684 players.
2025	**AL KHWARIZMI CTF**, Ranked **6th of 650+ teams** (national) as team DU_Restless_goblins.
2025	**Battle of Brains 2025**, Ranked **10th of 105** in the intra-university programming contest, University of Dhaka.
2024	**Battle of Brains 2024**, Ranked **14th of 68** in the intra-university programming contest, University of Dhaka.
2023	**DU CTF 2023**, Ranked **10th of 100+ teams** as team DU_CTF_NOOBS; received a prize at the Department of CSE, University of Dhaka.

Extracurricular Activities

Battle of Brains 2026, University of Dhaka

Organizer & Problem Setter *2026*

Organized an intra-university programming contest between freshers and sophomores and set the problems *Ask for the Deck If you Wish* and *Tumi, Ami ar Ak Cup Coffee, kobe!?*.

CSEDU Students' Club, University of Dhaka

Executive Member *February 2024 - November 2025*

Organized departmental events and collaborated with other DU departments on inter-department activities.

Code Samurai 2024

Volunteer *2024*

Volunteered at Code Samurai, the biggest inter-university hackathon in Bangladesh, organized by the University of Dhaka in collaboration with BJIT, Japan.

CAIL Workshop 2026

Participant *2026*

Hands-on workshop: *From Language Model to Service: LLM Inference, APIs and Scalable Deployment*.

Reference

Name Dr. Md. Mosaddek Khan

Designation Associate Professor

Institution Department of Computer Science and Engineering, University of Dhaka

Email mosaddek@du.ac.bd